M Tech (AIML)_Machine Learning _ Assignment 1

Problem Statement 1

In large-scale agri-supply chains, automated **visual inspection systems** are deployed to ensure product authenticity and grade consistency. One such challenge is the **morphological discrimination of visually similar dried fruit variants** using geometric descriptors extracted from imaging pipelines.

You are provided with a dataset derived from an image-processing system capturing **shape-based descriptors of dried fruit samples**. The task is to design a **robust, interpretable, and generalizable classification system** that can distinguish between two morphologically similar variants.

1. Exploratory Structural Analysis (ESA) [2 Marks]

a. Perform a **structural audit** of the dataset:

- Feature types, ranges, distributions
- Detect hidden patterns (e.g., scale imbalance)

b. Construct **multi-perspective visualizations**:

- Distribution plots (univariate)
- Pairwise relationships (bivariate)
- Class-conditional feature distributions

c. Compute and interpret **feature interaction structure**:

- Correlation matrix
- Mutual information (MANDATORY)

d. Explain why *correlation alone is insufficient* for feature selection in non-linear models.

2. Data Conditioning & Representation Learning [3 Marks]

a. Perform **data sanitization pipeline**:

- Outlier detection using **at least two methods** (e.g., IQR, Isolation Forest)
- Skewness correction (log/Box-Cox)

b. Apply **representation transformation techniques**:

- Standardization vs Normalization vs Robust Scaling
- Justify mathematically which is suitable and why

c. Perform **feature importance estimation using at least two paradigms**:

- Model-based (e.g., tree importance)
- Statistical (e.g., ANOVA / MI)

d. Construct at least **one derived feature** based on domain intuition (e.g., shape ratio) and justify it.

3. Learning Paradigm Design [5 Marks]

a. Perform **data partitioning strategies**:

- Hold-out split (80–20)
- Stratified K-Fold Cross Validation

Justify when stratification is critical.

b. Train and compare **at least THREE models**:

- Logistic Regression (baseline)
- Decision Tree
- One advanced model (choose from: Random Forest / SVM / Gradient Boosting)

---

c. Perform **hyperparameter optimization**:

- Grid Search or Random Search
- Clearly define search space and rationale

4. Model Evaluation & Generalization Analysis [3 Marks]

a. Evaluate models using:

- Accuracy, Precision, Recall, F1-score
- ROC-AUC (MANDATORY)

b. Perform **bias-variance analysis**:

- Compare training vs validation performance

c. Conduct **error analysis**:

- Identify misclassified samples
- Explain *why* they failed

5. Interpretability & Explainability [2 Marks]

a. Use at least one interpretability technique:

- SHAP / Feature importance / Coefficients

b. Answer:

- Which features dominate decision-making?
- Are decisions **physically meaningful**?

6. Research Extension [3 Marks]

Choose ANY ONE:

1. **Model Robustness Study**
   Add synthetic noise and analyze performance degradation
2. **Feature Ablation Study**
   Remove top features and observe impact
3. **Dimensionality Reduction**
   Apply PCA and compare performance vs original space

 # Submission Requirements

- **Jupyter Notebook with:**
  - **Clean code**
  - **Markdown explanations**
  - **Justifications (STRICTLY REQUIRED)**
- **No justification =** zero marks for that subpart

first column must include a section specifying individual contribution, for example:

- Group Member 1 name – 100%
- Group Member 2 name – 100%
- Group Member 3 name – 0%
- Group Member 4 name – 100%
- Group Member 5 name – 100%